linear 1^{st} order ODE

$$\boxed{\underline{a_1(x)}\frac{dy}{dx} + \underline{a_0(x)}\,y = \underline{b(x)}}$$

Ex: $x^2 - (\cos x)y = (\sin x)\frac{dy}{dx}$

$(\sin x)\frac{dy}{dx} + (\cos x)y = x^2$ ← linear

$\frac{dy}{dx} + (\sin x)y^3 = e^x + 1$ ← not linear

$x^3 - y\frac{dy}{dx} = \tan x$ ← not linear

Ex:

$$x^2\frac{dy}{dx} + \underset{\substack{\parallel \\ (x^2)'}}{2x}\,y = \cos x$$

$$(x^2 \cdot y)' = \cos x$$

$$x^2 \cdot y = \sin x + C$$

$$y = \frac{\sin x}{x^2} + \frac{C}{x^2}$$

Example 1 Find the general solution to

(9) $$\frac{1}{x}\frac{dy}{dx} - \frac{2y}{x^2} = x\cos x\,, \qquad x > 0\,.$$

$\cdot\frac{1}{x}$ ← integrating factor

$$\frac{1}{x^2}\frac{dy}{dx} - \frac{2}{x^3}y = \cos x$$

$$\overset{\shortparallel}{\left(\frac{1}{x^2}\right)'} \Rightarrow \left(\frac{1}{x^2}\cdot y\right)' = \cos x$$

$$\frac{1}{x^2}\cdot y = \sin x + C$$

$$\boxed{y = x^2\sin x + Cx^2}$$

Finding integrating factor

Find $\mu(x) \rightarrow$ such that

$$\frac{dy}{dx} + P(x)y = Q(x) \leftarrow \text{standard form}$$

$$\mu(x)\frac{dy}{dx} + \underbrace{\mu(x)P(x)}_{\mu'}y = \mu(x)Q(x)$$

$$\mu' = \mu \cdot P \leftarrow \text{separable} \qquad \frac{d\mu}{dx} = \mu P(x)$$

$$\Rightarrow \int \frac{d\mu}{\mu} = \int P(x)\,dx \Rightarrow \ln \mu = \int P(x)\,dx + \cancel{c} \quad (c=0)$$

$$\Rightarrow \boxed{\mu(x) = e^{\int P(x)\,dx}}$$

Method for Solving Linear Equations

(a) Write the equation in the standard form

$$\frac{dy}{dx} + P(x)y = Q(x) \ .$$

(b) Calculate the integrating factor $\mu(x)$ by the formula

$$\mu(x) = \exp\left[\int P(x)\,dx\right] \ .$$

(c) Multiply the equation in standard form by $\mu(x)$ and, recalling that the left-hand side is just $\frac{d}{dx}\left[\mu(x)y\right]$, obtain

$$\underbrace{\mu(x)\frac{dy}{dx} + P(x)\mu(x)y}_{\frac{d}{dx}\left[\mu(x)y\right]} = \mu(x)Q(x) \ ,$$

$$\frac{d}{dx}\left[\mu(x)y\right] = \mu(x)Q(x) \ .$$

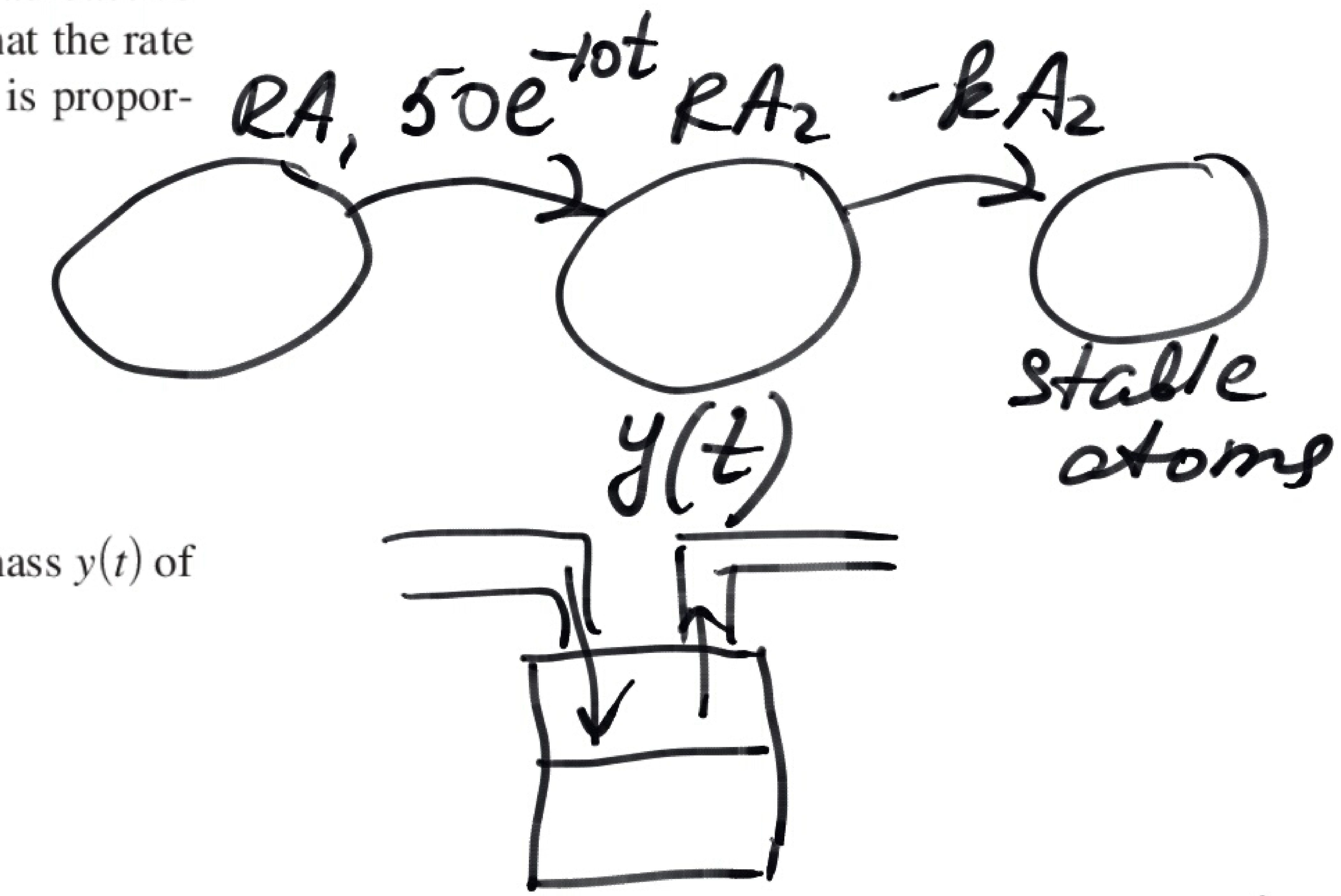
(d) Integrate the last equation and solve for y by dividing by $\mu(x)$ to obtain (8).

Example 2 A rock contains two radioactive isotopes, RA_1 and RA_2, that belong to the same radioactive series; that is, RA_1 decays into RA_2, which then decays into stable atoms. Assume that the rate at which RA_1 decays into RA_2 is $50e^{-10t}$ kg/sec. Because the rate of decay of RA_2 is proportional to the mass $y(t)$ of RA_2 present, the rate of change in RA_2 is

$$\frac{dy}{dt} = \text{rate of creation} - \text{rate of decay} ,$$

$$\boxed{\frac{dy}{dt} = 50e^{-10t} - ky ,} \tag{12}$$

not separable
is linear



where $k > 0$ is the decay constant. If $k = 2$/sec and initially $y(0) = 40$ kg, find the mass $y(t)$ of RA_2 for $t \geq 0$.

$$\frac{dy}{dt} + 2y = 50e^{-10t}$$

$$\mu(t) = e^{\int 2dt} = e^{2t}$$

$$e^{2t}\frac{dy}{dt} + \underbrace{2e^{2t}}_{(e^{2t})'}y = 50e^{-8t}$$

$$(e^{2t}\cdot y)' = 50e^{-8t}$$

$$e^{2t}\cdot y = -\frac{50}{8}e^{-8t} + C$$

$$y = -\frac{25}{4}e^{-10t} + Ce^{-2t}$$

$$40 = -\frac{25}{4} + C$$

$$C = 40 + \frac{25}{4} = \frac{185}{4}$$

$$\boxed{y = -\frac{25}{4}e^{-10t} + \frac{185}{4}e^{-2t}}$$